



More complex binomial products

Task A: Using equivalent binomials

1) Use the area models to expand and simplify these expressions.

a)

	$-y$	7	
y	$-y^2$	$7y$	
4	$-4y$	28	

$$\begin{aligned}
 &(y + 4)(7 - y) \\
 &(y + 4)(-y + 7) \\
 &\equiv -y^2 + 3y + 28
 \end{aligned}$$

c)

	y	5	
$-y$	$-y^2$	$-5y$	
15	$15y$	75	

$$\begin{aligned}
 &(15 - y)(y + 5) \\
 &(-y + 15)(y + 5) \\
 &\equiv -y^2 + 10y + 75
 \end{aligned}$$

b)

	y	-8	
y	y^2	$-8y$	
-8	$-8y$	64	

$$\begin{aligned}
 &(-8 + y)(y - 8) \\
 &(y - 8)(y - 8) \\
 &\equiv y^2 - 16y + 64
 \end{aligned}$$

d)

	-10	$-y$	
x	$-10x$	$-xy$	
$-y$	$10y$	y^2	

$$\begin{aligned}
 &(x - y)(-y - 10) \\
 &(x - y)(-10 - y) \\
 &\equiv -10x + 10y - xy + y^2 \\
 &\text{Terms in any order}
 \end{aligned}$$

2) Which of these expressions can be written as the difference of two squares.

a) $(x + 9)(9 - x)$
 $\equiv (9 + x)(9 - x)$

d) $(9 + x)(x - 9)$
 $\equiv (x + 9)(x - 9)$

g) $(-x + y)(y - x)$

b) $(x - 9)(9 - x)$

e) $(x + y)(y + x)$

h) $(x - y)(y + x)$
 $\equiv (x - y)(x + y)$

c) $(x + 9)(-x - 9)$

f) $(x + y)(-x + y)$
 $\equiv (y + x)(y - x)$

i) $(x - y)(y - x)$

3) Which products of two binomials give the expressions in the box?

a) $(x + 9)(9 - x)$
 $\equiv (9 + x)(9 - x)$
 $\equiv 81 - x^2$

d) $(9 + x)(x - 9)$
 $\equiv (x + 9)(x - 9)$
 $\equiv x^2 - 81$

h) $(x - y)(y + x)$
 $\equiv (x - y)(x + y)$
 $\equiv x^2 - y^2$

f) $(-x + y)(x + y)$
 $\equiv (y - x)(y + x)$
 $\equiv y^2 - x^2$

$x^2 - 81$

$81 - x^2$

$x^2 - y^2$

$y^2 - x^2$



Task B: Coefficients other than 1

1) Expand and simplify each product of two binomials.

$$\begin{aligned} \text{a)} \quad (2x - 6)(x + 3) \\ &\equiv 2x^2 + 6x - 6x - 18 \\ &\equiv 2x^2 - 18 \end{aligned}$$

$$\begin{aligned} \text{c)} \quad (3x - 6)(2x + 6) \\ &\equiv 6x^2 + 18x - 12x - 36 \\ &\equiv 6x^2 + 6x - 36 \end{aligned}$$

$$\begin{aligned} \text{e)} \quad (3x - 6)(3x - 6) \\ &\equiv 9x^2 - 18x - 18x + 36 \\ &\equiv 9x^2 - 36x + 36 \end{aligned}$$

$$\begin{aligned} \text{b)} \quad (3x - 6)(x + 3) \\ &\equiv 3x^2 + 9x - 6x - 18 \\ &\equiv 3x^2 + 3x - 18 \end{aligned}$$

$$\begin{aligned} \text{d)} \quad (3x - 6)(2x - 6) \\ &\equiv 6x^2 - 18x - 12x + 36 \\ &\equiv 6x^2 - 30x + 36 \end{aligned}$$

$$\begin{aligned} \text{f)} \quad (3x - 6)(3x + 6) \\ &\equiv 9x^2 + 18x - 18x - 36 \\ &\equiv 9x^2 - 36 \end{aligned}$$

2) What do you notice about your answer to a) and f)? Can you explain why this happens.

Two of the partial products are zero pairs so the final expression only has two terms.

f) is clearly the difference of two squares as both binomials have the term $3x$ and the other terms -6 and 6 are a zero pair.

The first binomial in a) $(2x - 6)$ can actually be written as $2(x - 3)$ so the product is the same as $2(x - 3)(x + 3)$ Which is the difference of two squares all multiplied by 2. $2(x^2 - 3^2) \equiv 2(x^2 - 9) \equiv 2x^2 - 18$

3) Expand and simplify each product of two binomials.

$$\begin{aligned} \text{a)} \quad (2x + 3y)(4x + 5y) \\ &\equiv 8x^2 + 10xy + 12xy + 15y^2 \\ &\equiv 8x^2 + 22xy + 15y^2 \end{aligned}$$

$$\begin{aligned} \text{c)} \quad (3y - 4x)(2x - 5y) \\ &\equiv 6xy - 15y^2 - 8x^2 + 20xy \\ &\equiv -8x^2 + 26xy - 15y^2 \end{aligned}$$

$$\begin{aligned} \text{e)} \quad (-5y + 2x)(3y - 4x) \\ &\equiv -15y^2 + 20xy + 6xy - 8x^2 \\ &\equiv -8x^2 + 26xy - 15y^2 \end{aligned}$$

$$\begin{aligned} \text{b)} \quad (2x + 5y)(4x + 3y) \\ &\equiv 8x^2 + 6xy + 20xy + 15y^2 \\ &\equiv 8x^2 + 26xy + 15y^2 \end{aligned}$$

$$\begin{aligned} \text{d)} \quad (3x - 4y)(5y - 2x) \\ &\equiv 15xy - 6x^2 - 20y^2 + 8xy \\ &\equiv -6x^2 + 23xy - 20y^2 \end{aligned}$$

$$\begin{aligned} \text{f)} \quad (5y + 2x)(-3y - 4x) \\ &\equiv -15y^2 - 20xy - 6xy - 8x^2 \\ &\equiv -8x^2 - 26xy - 15y^2 \end{aligned}$$

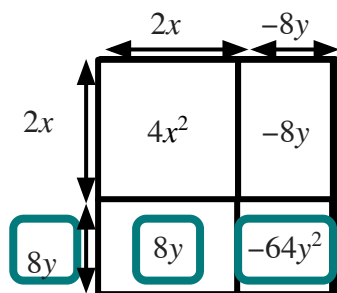
4) What do you notice about your answer to c) and e)? Can you explain why this happens.

They are the same because $(2x - 5y) \equiv (-5y + 2x)$



5) Izzy wants to find the expanded form of the product $(2x - 8y)^2$

a) Spot and correct any mistakes she has made.



$$\equiv 4x^2 - 64y^2$$

The binomial on the height of the square should have a term of negative 8. The partial products on the bottom row should be $-8y$ and $64y^2$



6) Can you write two binomials which will have the product $4x^2 - 64y^2$

Is there more than one way? $(2x - 8y)(2x + 8y)$ or $(-2x - 8y)(-2x + 8y)$ and any rearrangement of terms within the binomials.

